



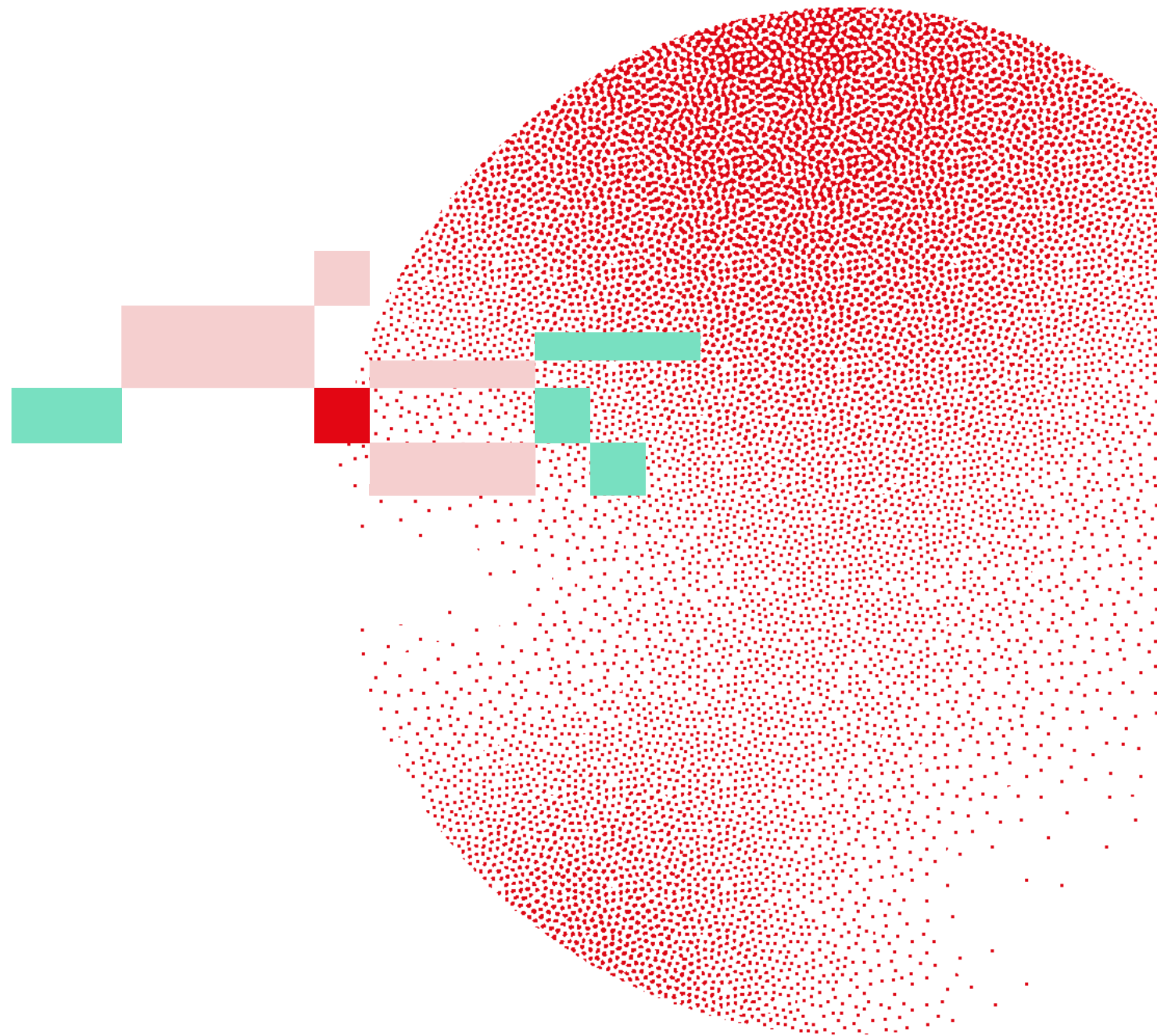
Swiss Institute of
Bioinformatics

SINGLE-CELL TRANSCRIPTOMICS WITH R

Introduction

Deepak Tanwar

March 18-20, 2026



All SIB courses abide the ELIXIR Code of Conduct

We **value** each other's perspectives providing a safe environment for people to be themselves.

We will **maintain** high ethical standards across all ELIXIR events.

We **adopt** a zero-tolerance approach to harassment and discrimination in any form.

We will **apply** honesty and integrity in the dealing of any transgressions against the Code.

We are **committed** to making ELIXIR events a collaborative, supportive and enjoyable experience.

We will **ensure** that our environment allows everyone to feel respected and included.

Trainers/ organisers/ helpers

»» Deepak Tanwar

- Trainer at SIB & Bioinformatics Specialist at UZH



»» Joana Carlevaro-Fita

- Trainer at SIB & Senior Bioinformatician at UNIBE



»» Nikolai Püllen

- Single Cell Expert Genomics at FGCZ



»» Geert van Geest

- Training Project Manager



»» Patricia Palagi

- Training Group Leader at SIB



Learning outcomes

- ⌘ Explain the **characteristics** of the most-used **methods** to scRNA-seq data
- ⌘ Perform **quality control** at different steps of the analysis
- ⌘ Use **dimensionality reduction** (PCA, t-SNE, UMAP)
- ⌘ Apply **best practices** for **translating** single cell transcriptomic data into **biological knowledge**

Learning experiences

- » Lectures
- » Quizzes
- » Exercises

Communication

- » Course website: <https://sib-swiss.github.io/single-cell-r-training>
- » Slack:
 - » [#all-single-cell-transcriptomics-march-2026](#)
 - » [#q-and-a](#)
 - » [#projects{1.....6}](#)
- » Shared document
 - » General links for the course
 - » Please write your name for the exam

Course etiquette



Video on when
possible

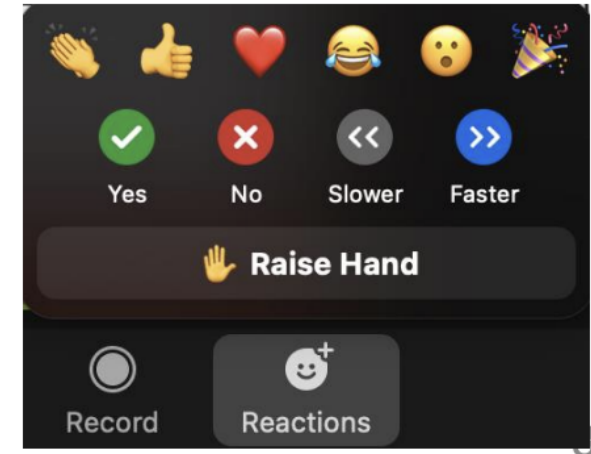
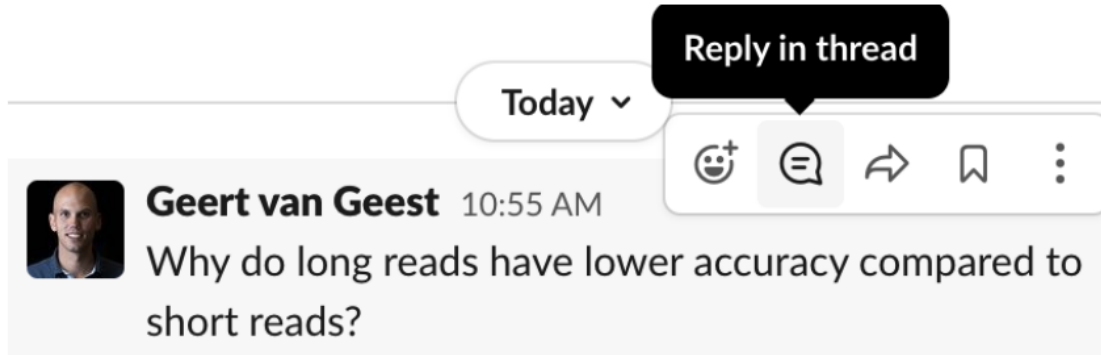


Mute when
not speaking

Use your **first** and **last** name as your Zoom ID
Right-click on yourself > Rename..

Asking questions

- During lectures: zoom functionality.
The Zoom chat will be ignored!
- During exercises: **#q-and-a** on slack



Quiz: 1-4

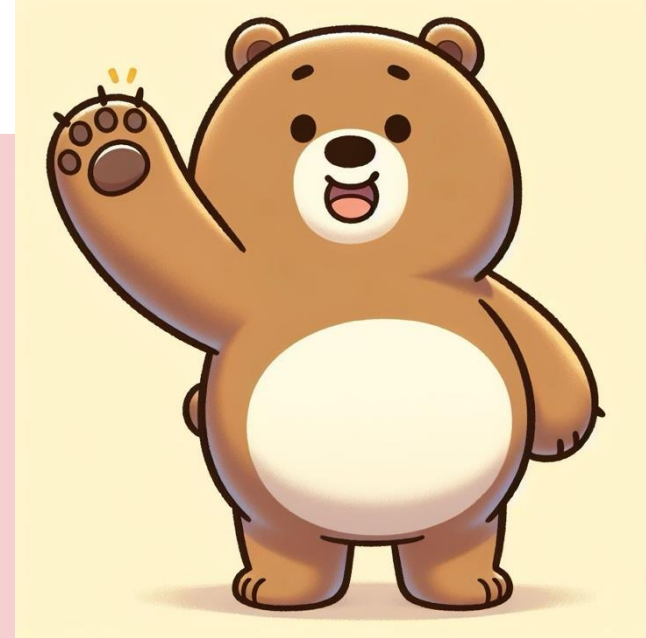
Introduction round

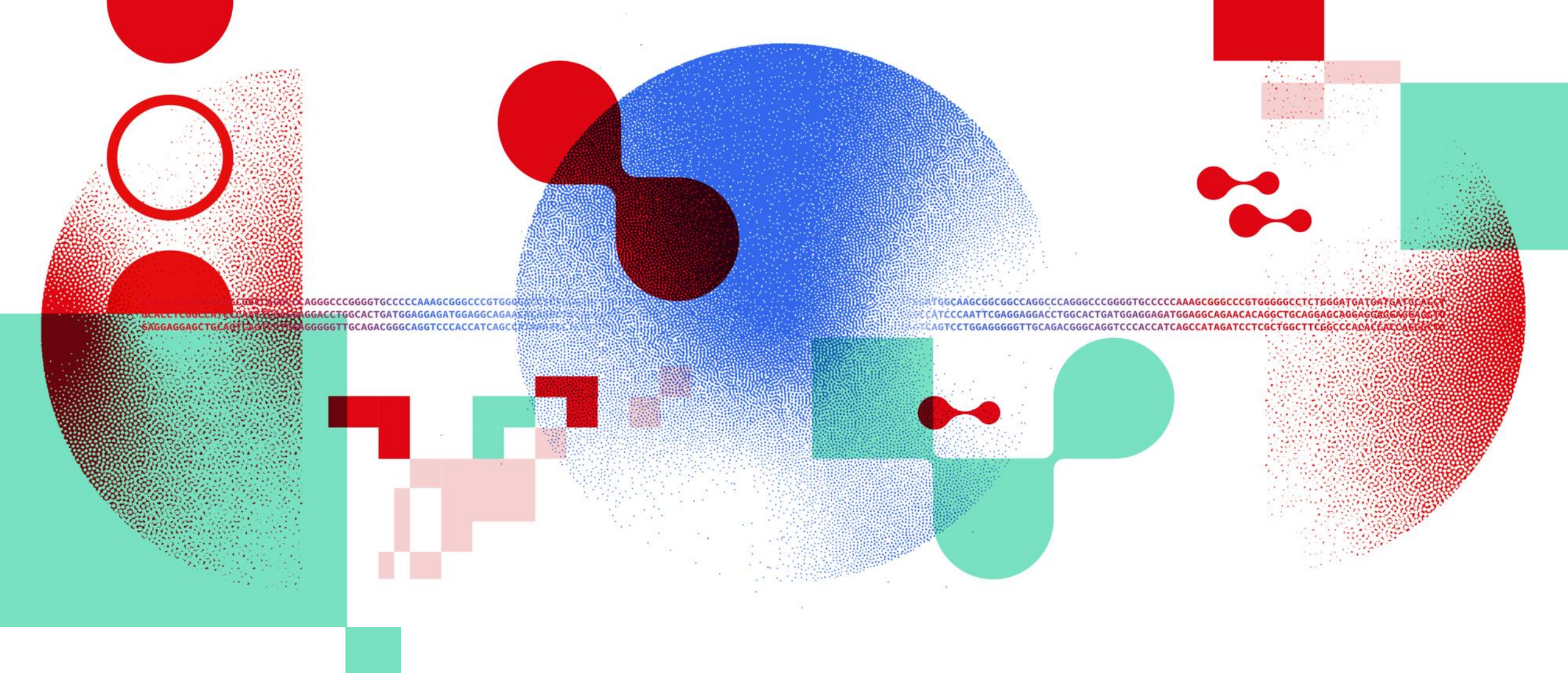
Your name

What do you do? (PhD/ postdoc)

Your topic of research.

Why are you joining the course?





Thank you

DATA SCIENTISTS FOR LIFE

sib.swiss